7

Equivalence relations

A first look at relations more abstractly. Here we will look at properties some types of relationship have, and observe that not many things satisfy those properties. This is to motivate the more relaxed axioms for a category. We will now start using more formal notation.

Category theory is based on the idea of relationships. We build contexts in which to study things. We build them out of objects, and relationships between objects.

Although this is quite general, we do still have to be a *bit* specific about the relationships, because if we allow any old type of relationship we might get complete chaos. So we're going to put a few mild conditions on the type of relationship we study. In this chapter we're going to explore the idea of putting conditions on relationships, and then in the next chapter we'll look at what conditions we actually use in category theory.

In math, and in life, we often start out with conditions that are rather stringent, either to be on the safe side or because we're conservative, or unimaginative, or exclusive. For example, at some egregious point in history, marriage had to be between two people of the same race. Now that condition has mostly been dropped, and in some places we've agreed that the two people don't have to be different genders any more. I believe that thinking clearly, from first principles, leads to less reliance on fearful boundaries, and greater inclusivity.

In this chapter we're going to explore a type of relationship that has rather stringent conditions on it, before seeing how we can relax the conditions but still retain organized structure when we define categories. The idea is to do with generalizing the notion of equality.

7.1 Exploring equality

If we are going to use a more relaxed and inclusive notion of equality, what sorts of things should we look for that won't make our arguments and deductions fall apart too much?

Let's think about how we use equality in building arguments. One of the key ways we use it is in building a long string of small steps as shown on the right.

$$\begin{array}{rcl} a & = & b \\ b & = & c \\ c & = & d \\ d & = & e \\ \boxed{\text{so } a} & \boxed{=} & \boxed{e} \end{array}$$

Suppose now we say that one person is "about the same age as" another if they were born within a month. We could write this as $A \overset{\text{month}}{=\!=} B$ if the people involved are our usual "person A" and "person B". Now we can try making a long string of these, as shown. But we get a problem.

$$\begin{array}{rcl} A & \overset{\text{month}}{=\!=} & B \\ B & \overset{\text{month}}{=\!=} & C \\ C & \overset{\text{month}}{=\!=} & D \\ D & \overset{\text{month}}{=\!=} & E \end{array}$$

Here's the issue: can we now deduce that $A \overset{\text{month}}{=\!=} E$? We can't. It *might* be true, but it doesn't follow by logic, because A and E could be almost four months apart. If we keep going with enough people, we could get the people at the far ends to be any number of years apart. That makes for an unwieldy situation we might not want to deal with.

Things To Think About

T 7.1 Can you think of a way of saying two numbers are "more or less the same" which avoids this problem? Does it result in some other problems? We will come back to this in Section 7.7.

Another property of equality we habitually use is that it doesn't matter which way round we write it: $a = b$ and $b = a$ mean the same thing. This is not true for inequalities like $a < b$, so manipulating those takes more concentration as we really have to pay attention to which side is which.

For example if $a = b$ then we can immediately deduce that $a - b = 0$ and also $b - a = 0$. However, if instead we start with $a < b$ then we get $a - b < 0$ but $b - a > 0$. It takes some thought to get those the right way round.

Finally it might never have occurred to you to think about the fact that everything equals itself. But if anything failed to equal itself then all sorts of weird things would happen.[†]

7.2 The idea of abstract relations

We're now going to look at how we express conditions on a relationship without knowing what the relationship is yet. This involves going a level more

[†] In fact everything would probably equal 0 in which case everything *would* equal itself, we'd have a contradiction, and the world would implode.

abstract, not just on the objects (which we might call a, b, c or A, B, C) but on the relationship itself.

If we're talking about numbers we might think about inequality relationships such as $1 < 5$ and $6 < 100$. We might then refer abstractly to $a < b$, without specifying particular numbers a and b. When we take it a step further we might be thinking about various different relationships, like $a < b$ or $a = b$, and want to refer to a general relationship without specifying the relationship or the numbers. We might write it as "$a \, R \, b$" or $a \mathrel{\square} b$. I like the R because it is asymmetrical, reminding us that the left- and the right-hand sides might be playing different roles here, as with <. But I like the notation with the box because it reminds us that we can put something in the box. So I will make the box asymmetrical like this: $a \mathrel{\sqsubset\!\!\supset} b$.

Technically we could call this a *binary* relation because it's a relation that involves only two things at a time. But I will typically say "relation" and understand it to be binary; if we ever considered relations between more than two objects we could make that clear at that point.

Then the idea is we start with some world that we're going to study, say the world of people, or numbers, or days of the week, or shapes. In the above example about people's ages I sort of implicitly only considered applying the relationship to people. When we're talking about abstract relationships in math we should explicitly declare what world we're considering, so that we don't try and consider the relationship beyond the context intended. So we might say: consider the set of real numbers, and the relation $\sqsubset\!\!\supset$ defined on it by

$$\boxed{a \mathrel{\sqsubset\!\!\supset} b \quad \text{whenever} \quad a < b}$$

It's important to note that "$a \mathrel{\sqsubset\!\!\supset} b$" is a statement (with a verb in it) which might turn out to be true or might turn out to be false. This is very different from when we put a binary operation in between two things. For example $a + b$ is not a statement — it has no verb. If a and b are numbers then $a + b$ is just another number.

We are now going to examine what properties relations might have. In all of what follows we start by fixing a world, that is, a set of objects S, and a relation $\sqsubset\!\!\supset$ that we are considering in that world.

7.3 Reflexivity

A relation $\sqsubset\!\!\supset$ is called *reflexive* if it sees everything to be related to itself. That is, every object a satisfies $a \mathrel{\sqsubset\!\!\supset} a$. It's important to be clear that this is a property of the relation $\sqsubset\!\!\supset$, not a property of any particular object a. The condition of reflexivity only holds for the relation $\square$ if the condition $a \square a$ holds for all objects a in S.

At this point I want to introduce another piece of notation because I'm already bored of typing "for all". Mathematicians use this upside-down A to mean "for all": $\forall$. We will also continue to use the symbol $\in$ to indicate that something is in a set. So here is the full definition of reflexivity:

Definition 7.1 A relation $\square$ on a set S is called *reflexive* if

$$\forall\, a \in S,\ a \square a.$$

Translation: "for all a in S, a is related to a".

Example 7.2 Let $S = \mathbb{R}$ (the real numbers) and let $\square$ be the relation $=$. Now, given $a \in \mathbb{R}$ certainly $a = a$, so $\square$ is indeed reflexive.

Note that when we're proving "for all" statements we don't have to sit and test the conditions on every single object in turn (which would not be possible in an infinite set anyway). What we can do instead is pick one arbitrary member of the set (so we use some sort of letter to represent it) and then test the conditions only using properties that are known to be shared by every member of the set. So if it's the set of all numbers, we pick a random one and make sure we don't use anything about it except that it's a number. If it's the set of all even numbers we can use the fact that it's divisible by 2. It might look like we've only tested one number, but this is one of the powerful aspects of abstraction and why we "turn numbers into letters" — we've only tested one number abstractly, but effectively we've tested every example.

Things To Think About

T 7.2 Is the relation $<$ on $\mathbb{R}$ reflexive?

Let $a \in \mathbb{R}$. Then it is *not* true that $a < a$, so $\square$ is *not* reflexive. This is an extreme example of non-reflexivity, in which *no* object is related to itself. However, in general in order to fail reflexivity we don't have to be that extreme. A "for all" statement fails even if it only fails in one place; it doesn't have to fail everywhere. The next example is less "extreme".

Things To Think About

T 7.3 Let S = the set of all people in the world, and let $\square$ be the relation "is in love with". Is $\square$ reflexive?

The question we have to ask ourselves is: if we take any random person a in the world, are they necessarily in love with themselves? Well, we can't tell for sure. Some people are in love with themselves but some people aren't. This means that $\boxed{}$ is *not* reflexive because there exists at least one $a \in S$ such that $a \not\boxed{}\, a$. I'll use this notation of a slanted line through the box to mean that the relation does not hold in this case. Mathematicians also use this backwards E to say "there exists": $\exists$. "For all" and "there exists" are called *quantifiers* in logic, because they quantify (specify) the scope of a logical statement, that is, they tell us something about where the statement holds.

Anyway we can formally write down what we've found like this:

$$\exists\, a \in S \ \text{ such that } \ a \not\boxed{}\, a$$

Translation: "there exists a in S such that a is not related to a".

This shows that this particular $\boxed{}$ is not reflexive. If there's one case that contradicts a "for all" statement it's called a *counterexample*. There might well be more than one counterexample, but you only need one in order to show that a "for all" statement isn't true.

Things To Think About

T 7.4 Are these relations reflexive? In each case take the set S to be the set of all people in the world.

1. $\boxed{}$ = "is the same height as"
2. $\boxed{}$ = "is taller than"
3. $\boxed{}$ = "is at school with"

If it confuses you how to use "is the same height as" as a relation, imagine putting it inside the box $\boxed{}$ everywhere there is a box in the definition. To investigate reflexivity we need to see if this is true:

$$\forall a \in S \quad a \,\boxed{\text{is the same height as}}\, a$$

Translation: "for all people a in the world, a is the same height as a".

This is true, so the relation is indeed reflexive.

For "is taller than" we need to see if this is true:

$$\forall a \in S \quad a \,\boxed{\text{is taller than}}\, a$$

This is another "extreme" failure: given any person a, a is definitely not taller than a (you can't be taller than yourself), so "is taller than" is not reflexive.

The last example, "is at school with" is more subtle. We need to check this:

$$\forall a \in S \quad a \,\boxed{\text{is at school with}}\, a$$

Now, if a person a is at school at all, then they are at school with a (they are at school with themselves). However, some people aren't at school. In math we usually like being completely clear by picking a specific example: say me. I am a person, and I am not at school, therefore I am not at school with myself. I am a counterexample showing that this relation is not reflexive.

We'll now go on to a slightly more complicated condition on relations.

7.4 Symmetry

A relation is called *symmetric* if it's "the same both ways round", that is, it doesn't matter which way round you write it. That doesn't mean it's always true both ways round, it just means that its truth doesn't change if you write it the other way. Here's the formal definition.

Definition 7.3 A relation $\sqsupset$ on a set S is called *symmetric* if

$$\forall\, a, b, \in S, \text{ if } a \sqsupset b \text{ then } b \sqsupset a$$

Translation: "for all a and b in S, if a is related to b then b is related to a".

Note that the definition says	$a \sqsupset b$	$\Longrightarrow$	$b \sqsupset a$
but if we switch the roles of a and b we get	$b \sqsupset a$	$\Longrightarrow$	$a \sqsupset b$
and so putting those together we have	$a \sqsupset b$	$\Longleftrightarrow$	$b \sqsupset a$

So for a symmetric relation the statements $a \sqsupset b$ and $b \sqsupset a$ are logically equivalent. The "symmetry" here is the fact that if we switch the positions of a and b then the statement doesn't change, logically speaking; it just looks different.

Let's try some of the same examples as before.

Example 7.4 Let $S = \mathbb{R}$ and let $\sqsupset$ be the relation $=$. Now let $a, b \in \mathbb{R}$. If $a = b$ then it is also true that $b = a$, so this relation is symmetric.

Note that we are not saying a actually is equal to b, we are *supposing* $a = b$ and seeing what we could deduce *if* that were true.

Example 7.5 Let $S = \mathbb{R}$ and let $\sqsupset$ be the relation $<$. Now let $a, b \in \mathbb{R}$. If $a < b$ then it is definitely not true that $b < a$, so this relation is not symmetric.

Note that we have again got ourselves a situation where we don't just have one counterexample, we have something that is *always* false, an "extreme" non-symmetric relation.† Here is an "in between" example.

Example 7.6 Let $S = \mathbb{Z}$ and let $\sqsupset$ be the relation "divides" (and recall

† In fact these are called "anti-symmetric".

that we write this as a vertical line). Now let $a, b \in \mathbb{Z}$. If $a \mid b$ then it is *possible* that $b \mid a$ (if they're equal) but it is not *necessarily* true: for example $2 \mid 4$ but $4 \nmid 2$.† So the relation is not symmetric.

In that example, there's a special case: when $a = b$ we do have $a \mid b$ and $b \mid a$. It might be tempting to think the relation "sometimes symmetric and sometimes not" but this is not how a "for all" condition works in math. If something is "sometimes true and sometimes not" then it's not *always* true, so the "for all" statement is false.

Things To Think About

T 7.5 Are these relations symmetric? In each case take the set S to be the set of people in the world.

1. $\square$ = "is the same height as"
2. $\square$ = "is the mother of"
3. $\square$ = "is a sister of"

In each case we need to check that the relevant thing is true *for all a* and b. We can start by letting a and b be any people, and then we put the given relation in the box $\square$ every time it appears in the definition of symmetry, and see whether the condition is necessarily true for all people.

The first relation is symmetric: if a is the same height as b then b must be the same height as a.

However the second one is not symmetric: if a is the mother of b then b definitely can't be the mother of a. Again, it has failed in an extreme way — it's not just that it's *sometimes* false the other way round, it's that it's *always* false the other way round.

The last one is another "in between" case, though it might seem symmetric at first sight. However if a is a sister of b then b *might* be a sister of a but might be a brother. This is perhaps a bit sneaky, but it means that this relation is not symmetric. If we said "sibling" then it would in fact be symmetric.

Note that reflexivity involved considering one object at a time, and symmetry involved considering two. Our last condition involves considering three.

7.5 Transitivity

Our last condition is about whether we can build up long chains of relations or not, a bit like when we built up long chains of equalities like the one shown on the right.

$$\begin{array}{llll} a = b & & & \\ & b = c & & \\ & & c = d & \\ & & & d = e \\ & \text{so } a = e & & \end{array}$$

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† We generally put a slanted line like this through things to express the fact that they're not true.

However, we only need to demand the condition for chains of length 2 (involving 3 objects), because from there we can build up longer ones if we want.

Definition 7.7 A relation $\sqsupset$ on a set S is called *transitive* if

$$\forall a, b, c \in S : \quad \Big\{ \; a \sqsupset b \text{ and } b \sqsupset c \; \Big\} \implies a \sqsupset c.$$

Translation: "for all a, b, c in S,
if a is related to b and b is related to c then a is related to c".

I've put those big curly brackets in to try and make it a bit clearer where the logic is, because it can be a bit confusing. Perhaps it would be clearer to do it more geometrically

like this:

$$\begin{array}{lll} a \sqsupset b & & \\ & b \sqsupset c & \\ a & \sqsupset & c \end{array}$$

or maybe like this:

$$\begin{array}{l} a \sqsupset b \sqsupset c \\ a \quad \sqsupset \quad c \end{array}$$

which I hope reminds you of this:

$$\begin{tikzcd} a \arrow[r] \arrow[rr, bend right] & b \arrow[r] & c \end{tikzcd}$$

Let's try some examples, starting with our usual motivating one, equality.

Example 7.8 Let $S = \mathbb{R}$ and let $\sqsupset$ be the relation $=$. Now let $a, b, c \in \mathbb{R}$. If $a = b$ and $b = c$ then it follows that $a = c$, so this relation is transitive.

Again, remember we're not saying that a, b and c actually *are* equal, we are just supposing that $a = b$ and also that $b = c$ to see what we can deduce.

Example 7.9 Let $S = \mathbb{R}$ and let $\sqsupset$ be the relation $<$. Now let $a, b, c \in \mathbb{R}$. If $a < b$ and $b < c$ then it follows that $a < c$, so this relation is transitive.

Now let's try some non-examples.

Example 7.10 If a is the mother of b and b is the mother of c then a is definitely *not* the mother of c, so the relation "is the mother of" is not transitive.

That was the extreme failure, where the outcome can never be true. Here's a subtly "in between" one.

Things To Think About

T 7.6 Suppose a, b, c are people, and a is a sister of b and b is a sister of c. Can you work out a slightly sneaky way in which a might not be a sister of c? Remember that a, b and c aren't necessarily *different* people.

Here is a counterexample. Consider this a, b and c:

a	$=$	me
b	$=$	my sister
c	$=$	me

So $a \square b$
and $b \square c$
but $a \not\square c$ as I am not my own sister.

This might make you feel a bit tricked, and I admit it is a little sneaky. But it's logically completely sound, and this is the sort of thing we need to be very careful about in math. In some cases the relation itself precludes the possibility of a and c being the same person, once we've declared that $a \square b$ and $b \square c$ (for example if the relation is "is the mother of") but that's because of the definition of $\square$ in that particular case.

Note that where reflexivity involves only one object at a time, symmetry involves two objects at a time, and transitivity involves three. And let me just re-emphasize that these are not necessarily three *distinct* objects.

Things To Think About

T 7.7 Are these relations transitive? In each case take the set S to be the set of people in the world.

1. $\square$ = "is the same height as"
2. $\square$ = "is married to"
3. $\square$ = "is in a choir with"

In each case we start by letting a, b and c be any three people (not necessarily distinct), and then we put the given relation in the box $\square$ every time it appears in the definition of transitivity, and see whether the statement is necessarily true for *all* people.

If a is the same height as b and b is the same height as c then yes, a is necessarily the same height as c, so this relation is transitive.

Now suppose a is married to b and b is married to c. According to the laws of most countries, the only way for this to be legal is if a and c are the same person, because otherwise b would be committing bigamy. In any case it is certainly *possible* for $a \square c$ to fail: take a and b to be any married couple, and then take $c = a$. Then a is not married to c.†

The third example shows us what would have happened in the previous example if bigamy were allowed, because it is in fact possible to be in a choir with more than one person (in fact I think by definition it's probably not a choir otherwise). However a could be in a choir with b, and b could be in a different choir with c, and so it doesn't necessarily follow that a and c are in a choir together. So this fails transitivity for a slightly different reason.

† Actually the very precise mathematician brain in me observes that some people have started marrying themselves as a radical act of self-love (or something) but there are plenty of people who haven't, whom we can pick for our counterexample.

Note that we can build up longer strings of relations using transitivity even though we only specified it for sticking two relations together. For example if we have the chain of individual relations on the right, we can use transitivity one step at a time to build up to $a \square e$.

$$\begin{array}{llll} a \square b & & & \\ & b \square c & & \\ & & c \square d & \\ & & & d \square e \end{array}$$

This is why we don't need to specify conditions on any more than two steps of the relation — that is, three objects — at a time.

7.6 Equivalence relations

We have looked at three conditions on relations that make them well-behaved in a similar way to the familiar relation =. If a type of relation satisfies all three then it is particularly well-behaved, and we name it in the following definition.

Definition 7.11 If a relation $\square$ on a set S is reflexive, symmetric and transitive then we call it an *equivalence relation*.

Things To Think About

T 7.8 Are the following relations equivalence relations? Here again S is the set of all people in the world.

1. $\square$ = "has the same birthday as"
2. $\square$ = "is older than"
3. $\square$ = "is a friend of"

You might wonder whether I mean "same birthday" as in the day of the year, or including the actual year of birth as well. In fact it doesn't matter, as long as you're consistent: as long as you pick one version and stick to it, this is an equivalence relation.

The big clue here is the words "the same". This indicates that the relation is based on =, it's just that we've used a process to turn our objects into something where = makes sense in an unambiguous way. Once we've done it, we know that = is an equivalence relation, and so our relation will also be one. It's a form of transferring structure that we'll see more of later.

By contrast "is older than" is not an equivalence relation. It is not reflexive, as nobody is older than themselves, and it is not symmetric, because if a is older than b then it's definitely not true the other way round. But it is transitive.

For the case of friendship things become a bit more philosophical, or perhaps, depressing.

Reflexivity: Let a be any person. Is it necessarily true that a is friends with themselves? I don't think so. Many people are not kind to themselves. Although some people are friends with themselves, there exist people who aren't, so I would say this relation is not reflexive.

Symmetry: Let a and b be any people. If a is a friend of b, is it true that b is a friend of a? I would say not: a might be a very good friend to b who treats them horribly in return.

Transitivity: This is probably the most obvious and least depressing. If a is a friend of b and b is a friend of c then a and c might not even know each other, so they certainly need not be friends, so this is not transitive.

This is obviously not a rigorous mathematical situation and is somewhat down to interpretation. However, the unambiguous part is how we translate the definitions of reflexive, symmetric and transitive into statements we can check in the real life situations; the ambiguity arises as the answers might not be clear-cut. Here is an example that isn't clear-cut but for different reasons.

Things To Think About

T 7.9 This time let S be the set of locations and $\square$ be the relation "is east of". Is this reflexive, symmetric, transitive?

Here it really depends whether we allow going all the way around the world as an option. Usually we'd think that if a is east of b then b has to be west of a and can't be east of it, but if we're on a globe perhaps everything is east of everything else? Unfortunately we often use rather silly terminology left over from an ignominious Eurocentric past in which there was "The East" and "The West", as if we still think the earth is flat.

Of course, if we did count going all the way round the world then everything would be east of everything, and this would reduce to being a very uninteresting relation in which everything is related to everything else. Instead we could restrict our "world" S somewhat, instead of considering all locations everywhere. We could restrict it to, say, one particular city, in which case it will be transitive but not reflexive or symmetric.

This shows that relations take on different characteristics depending on what context we're thinking about. Sometimes if something is behaving in an unwieldy way we put more conditions on it, but sometimes we can restrict our context instead. This is one of the many reasons it's important to be clear what context we're in at any point.

7.7 Examples from math

Let's look at some more mathematical examples now.

Things To Think About

T 7.10 Are the following relations equivalence relations? Here S is $\mathbb{Z}$, the set of integers.

1. $\leq$ ("less than or equal to")
2. divides
3. is congruent mod n (for some fixed n)

The first one $\leq$ is crucially different from just $<$ because of the "or equals" part. This means that it is now reflexive, because $a \leq a$. However this does not make it symmetric: for example $1 \leq 2$ but it is not true the other way round. The relation is transitive.

The relation "divides" is reflexive since everything divides itself, and we've already seen that it isn't symmetric. It is transitive because if $a \mid b$ and $b \mid c$ then it definitely follows that $a \mid c$. (You could try proving that, starting by making a formal definition of "divides".)

For congruence, remember that two integers a and b are called "congruent mod n" if $n \mid b - a$. I hope you *feel* that this relation ought to be an equivalence relation as we're talking about things being "the same" on an n-hour clock. A feeling isn't enough, but I think it's an important start.

Reflexivity: Given $a \in \mathbb{Z}$, $n \mid a - a$ as this is 0. So the relation is reflexive.

Symmetry: We need to show that if $a \equiv b$ then $b \equiv a$. Well, if $n \mid b - a$ then $n \mid a - b$ so this is also true.

Transitivity: We proved that this is true in Section 6.5.

So far the only equivalence relations we've seen have involved some sort of sameness. It's worth wondering how far we can push that.

Things To Think About

T 7.11 Can you make a definition of "more or less the same as" that is an equivalence relation? Why would it cause problems if it's not?

Earlier on we saw a version of "more or less the same as" that is not transitive, when we looked at people born within a month of each other. Although we did it for people and ages, that is essentially the same as for numbers. We could make "more or less the same as" mean "within ε of each other"† for any value of $\varepsilon > 0$ and it still wouldn't be transitive, no matter how small we made ε. (It would, however, be reflexive and symmetric.)

† Mathematicians traditionally use the Greek letter ε (epsilon) to represent unspecified small positive numbers.

One way of saying "more or less the same as" that is transitive is to round everything to the nearest whole number, or to a certain number of decimal places. This is how we take approximations of numbers, especially in physics, other experimental sciences, and in real life.

However, this results in other problems with things like 1.49 rounding to 1 but 1.5 rounding to 2 despite being almost the same as 1.49. It also means you can add two things that round to 0 and get something that does not round to 0, like when you somehow get full from eating a large quantity of negligible bites of something. We sometimes have to pick which problem to live with though, and usually the problem of anomalies with rounding is a bit irritating, where the problem of non-transitivity is catastrophic.

7.8 Interesting failures

There is plenty more to say about equivalence relations, but the last thing I want to draw attention to is how few things actually turn out to be equivalence relations. Moreover, many of the things that fail to be equivalence relations are mathematically interesting concepts that we may well still want to study. In math (and in life) if something doesn't fit our rigid constraints we can either continue to exclude them, or relax our constraints to let them in. In both math and life I believe in recognizing when our constraints have been unreasonably rigid, excluding things and people just because of preconceptions, out-dated practices, or fear. I believe in inclusivity in math and in life.

Abstract math has a reputation of being rigid and rule-based. It is true that any given context has its rules, but flexibility comes from continually finding ways to shift context or expand contexts in order to include concepts that previously didn't fit.

Equivalence relations are particularly well-behaved relations. But they are so well-behaved that they are, in a way, not that interesting. In fact all the ones we've seen are just some version of sameness where we map our world into a simpler one and apply sameness there, like having the same birthday, being the same height, being the same on an n-hour clock. Equivalence relations turn out to be logically the same as "partitions", where you just partition the world into sections and call two objects related if they're in the same section. Many interesting relations don't work like that, and in category theory we don't want to just exclude them. So we relax our conditions instead. This is what we're going to do in the next chapter.